Kinetic FanLab — Technology Statement (IBM Bob)

1. IBM Bob IDE Architecture & Full-Stack Engineering

Kinetic FanLab was architected, engineered, and validated using **IBM Bob IDE** in collaboration with **Gemini** under Team Lead **Troy Resendez**. Guided by strict architectural contracts (`AGENTS.md` and `Design.md`), IBM Bob governed a 6,300+ line Python and Streamlit production codebase (`app.py`) spanning collegiate (NCAA D1) and professional (NBA/NFL) experiences. Across the entire application, IBM Bob enforced zero-error **Pyright** static typing, modular session-state isolation across sports and teams, dynamic credential and schedule binding, and continuous automated test-suite verification.

2. Multi-Agent AI Orchestration, Signal Processing & IBM Cloud Stack

- **16-Agent Theatre Cast (IBM watsonx Orchestrate):** Sixteen declarative YAML agents deployed on IBM Cloud (`eu-de-0ec71755`) across four operational tiers (*Front Door*, *Front Stage*, *Backstage Crew*, *Back Door*) implement all **8 Official IBM Agentic AI Archetypes** with 2-for-1 redundancy, powering live conversational squad personas (`@coach`, `@conductor`, `@strategist`, `@scout`, and `@squad`).
- **Generative AI & Acoustic DSP (watsonx.ai): IBM Granite** (`granite-3-8b-instruct`) synthesizes real-time post-play narratives and localized stadium chants. Edge telemetry combines logarithmic SPL summation, a **Hampel 3-sigma outlier filter** (99.4% noise rejection), and a **50-seat BLE mesh** that cuts cellular backhaul by 98.0%.
- **Responsible AI & Governance (watsonx.governance):** Enforces a 21-pattern prompt-injection shield, 100% PII redaction, and verified regional dialect fairness (**DIR = 0.953–1.00**, EEOC four-fifths compliant).
- **Enterprise Persistence & Live Data Integration: IBM Cloudant NoSQL** (`us-south`) delivers sub-50ms hot state, **IBM Cloud Object Storage** (`eu-de`) archives immutable Parquet box scores with `SHA-256` receipts, and live **ESPN REST API** polling synchronizes real-time schedules and scores—verified under a 50-fan concurrency burst in **1.322 seconds** (26.44 ms P95 latency).